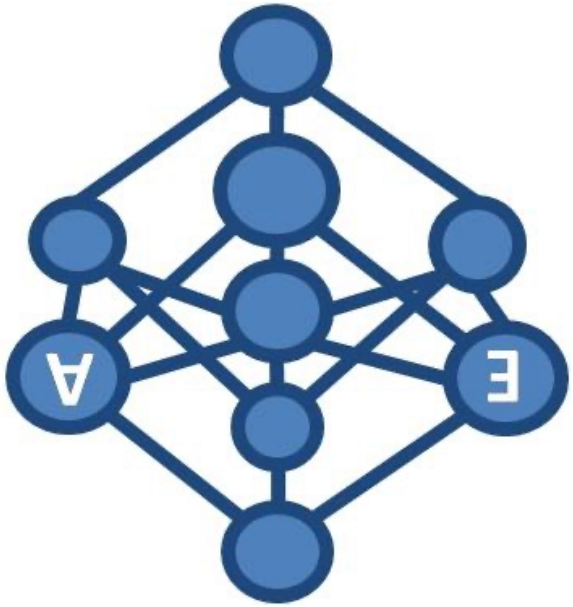
Probabilistic Graphical Models*

Bayesian Networks

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*Thanks to Carlos Guestrin, Pedro Domingos and many others for making their slides publically available

What you need to know from last class

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- Basic definitions of probabilities
- Independence
- Conditional independence
- The chain rule
- Bayes rule

Let‘s start on Bayesian Networks

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- One of the most exciting recent advancements in statistical AI

Judea Pearl won the ACM Turing Award 2012 for his fundamental contributions to artificial intelligence through the development of a calculus for probabilistic and causal reasoning



- Compact representation for exponentially-large probability distributions
- Fast marginalization
- Exploit conditional independencies

Representing Distributions by Enumeration

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- Consider $P(X_i)$
  - Assign a probability to each $x_i \in \mathrm{Val}(X_i)$
  - Number of parameters assuming $|\mathrm{Val}(X_i)| = k$?

$$k-1$$

- Now, consider $P(X_1,\ldots,X_n)$
  - How many parameters assuming again $|\mathrm{Val}(X_i)| = k$?
  - Same thing,

$$k^n-1$$

- **Bayesian networks will often need fewer parameters. What is the trick?**

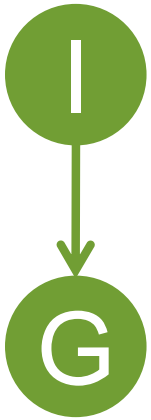


Conditional Parameterization (2 nodes)

- **Let us assume that**

  Grade is influenced by Intelligence

- Make use of **chain rule**



$$P(G,I) = P(G \mid I) \cdot P(I)$$

$$P(I = VH, G = B)$$

$$= P(I = VH) \cdot P(G = B \mid I = VH)$$

$$= 0.85 \cdot 0.1$$

$$= 0.085$$

	VH	H
P(I)	0.85	0.15

	I=	VH	H
P(G|I)	A	0.9	0.5
	B	0.1	0.5

Represent conditional distributions graphically

Same # of parameters as by enumeration

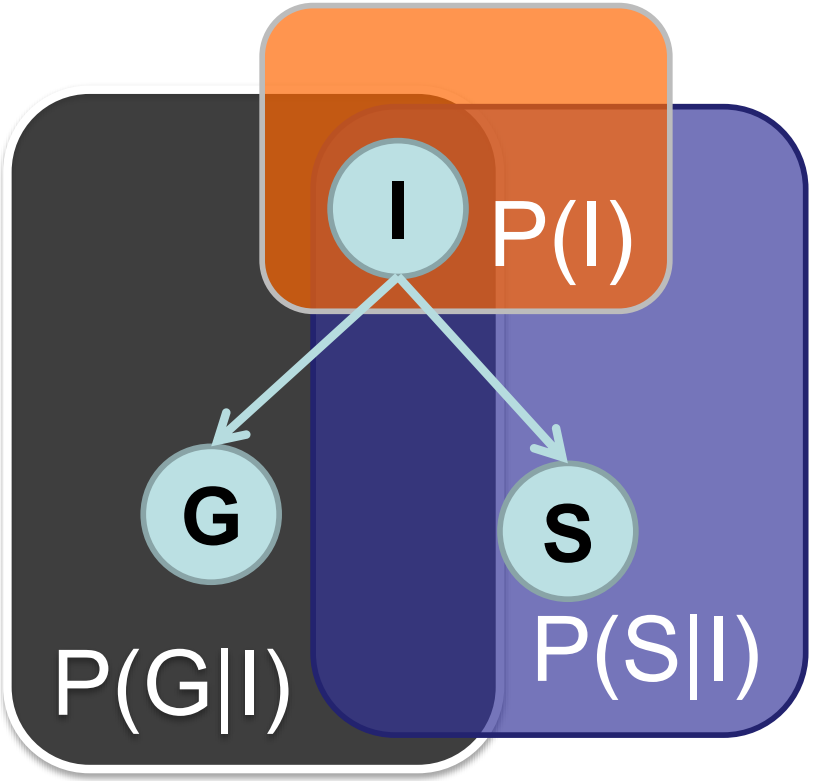
Conditional Parameterization (3 nodes) and what if variables are independent?

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- Grade and SAT score are influenced by Intelligence
- but $(G \perp S | I)$, i.e., $P(G | S, I) = P(G | I)$

$$
\begin{aligned}
& P(G, S, I) \\
&= P(G, S | I) \cdot P(I) \\
&= P(G | S, I) \cdot P(S | I) \cdot P(I) \\
&= P(G | I) \cdot P(S | I) \cdot P(I)
\end{aligned}
$$

Indpendence can lead to smaller # of parameters as by enumeration



Can we even get a linear complexity?

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- $\left(X_i \perp X_j\right), \forall i, j$ enough?

Can we even get a linear complexity?

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- $\left(X_i \perp X_j\right), \forall i, j$ enough?

Let ***X1 and X2*** *be drawn from* ***Bernoulli(p=0.5*** **)** [value TRUE with prob p and value FALSE with q=1-p] and ***X3 = X1 xor X2****.* Now, ***P(Xi,Xj) = 1/4 = P(Xi)P(Xj)***

X1	X2	X3	**P**
0	0	0	1/4
0	1	1	1/4
1	0	1	1/4
1	1	0	1/4

Since X3 is determined by X1 and X2, there are only four possible configurations (1 = TRUE, 0 = FALSE), each with probability 0.5 · 0.5 = 0.25. All others are zero due the XOR. So P(X3=1) = 0.25+0.25=0.5. So we have pairwise independency

Can we even get a linear complexity?

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Can we even get a linear complexity?

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- $(X_i \perp X_j), \forall i, j$ is not enough
- We must assume that $(\mathbf{X} \perp \mathbf{Y}), \forall \mathbf{X}, \mathbf{Y}$ subsets of $\{X_1, \ldots, X_n\}$

Let ***X1 and X2*** *be drawn from* ***Bernoulli(p=0.5*** **)** [value TRUE with prob p and value FALSE with q=1-p] and ***X3 = X1 xor X2****.* Now, ***P(Xi,Xj) = 1/4 = P(Xi)P(Xj).*** Since *X3 depends deterministically on X1 and X2 it cannot be the case that X1,X2 are independent of X3.*

- Now, we can write $P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i)$
- How many parameters?

$$n \cdot (k-1) = O(n)$$



Almost the whole PGM class in 22 slides

Let us now touch upon almost all topics of the lecture (representation, inference, and lerarning) so that you get the big picture and great motivation for the rest of the lecture already today.

For that use a simple Bayesian network for classifiction:

THE NAÏVE BAYES CLASSIFIER



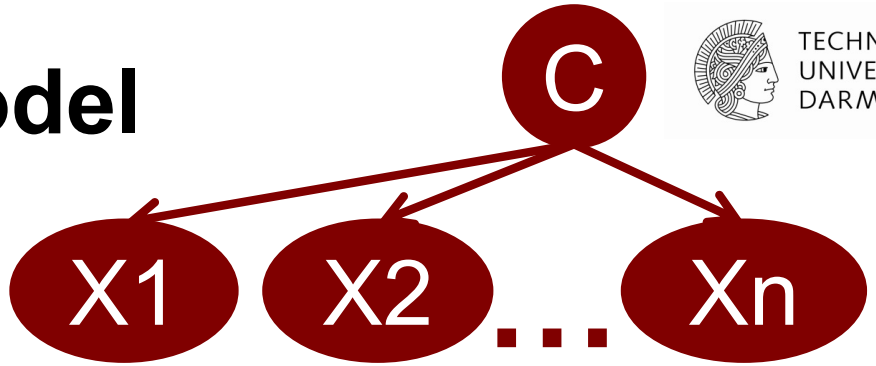
The Naïve Bayes Model

- **Your first real Bayesian network !**

- **Class variable:** C
- **Evidence variables:** $\{X_1,\ldots,X_n\}$
- **Assume** that $(\mathbf{X} \perp \mathbf{Y} \mid C), \forall \mathbf{X}, \mathbf{Y}$ subsets of $\{X_1,\ldots,X_n\}$

$$P(X_1,\ldots,X_n,C) = P(C)\cdot\prod_{i=1}^{n} P(X_i \mid C)$$

The Naïve Bayes Model



$$P(X_1,\ldots,X_n,C)$$

$$= P(C)\cdot P(X_1\,|\,C)\cdot P(X_2\,|\,X_1,C)\cdot\ldots\cdot P(X_n\,|\,X_1,\ldots X_{n-1},C)$$

So far, no assumptions. Just the chain rule.

Now, let us make the assumption that $\left(\mathbf{X}\perp\mathbf{Y}\,|\,C\right),\forall\mathbf{X},\mathbf{Y}$

Then we get the Naïve Bayes model

$$P(X_2\,|\,X_1,C)=P(X_2\,|\,C) \qquad P(X_n\,|\,X_1,\ldots X_{n-1},C)=P(X_n\,|\,C)$$

$$\left(X_1\perp X_2\,|\,C\right) \qquad \ldots \qquad \left(X_n\perp X_1X_2\ldots X_{n-1}\,|\,C\right)$$

Example application of Naïve Bayes: Spam Classification

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From: "" <takworlld@hotmail.com>
Subject: real estate is the only way... gem oalvgkay

Anyone can buy real estate with no money down

Stop paying rent TODAY !

There is no need to spend hundreds or even thousands for similar courses

I am 22 years old and I have already purchased 6 properties using the methods outlined in this truly INCREDIBLE ebook.

Change your life NOW !

==========

Click Below to order:
http://www.wholesaledaily.com/sales/nmd.htm

==========



Is this email spam or not?

Defintion of Classification

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Given

- A description of an instance, $x \in X$, where X is the *instance language* or *feature space*.
  - **How to represent text documents? E.g. Bag-Of-Words, i.e., we count how often each word of a fixed dictionary appears in the documents**
- A fixed set of categories: $C = \{c_1, c_2, \ldots, c_n\}$

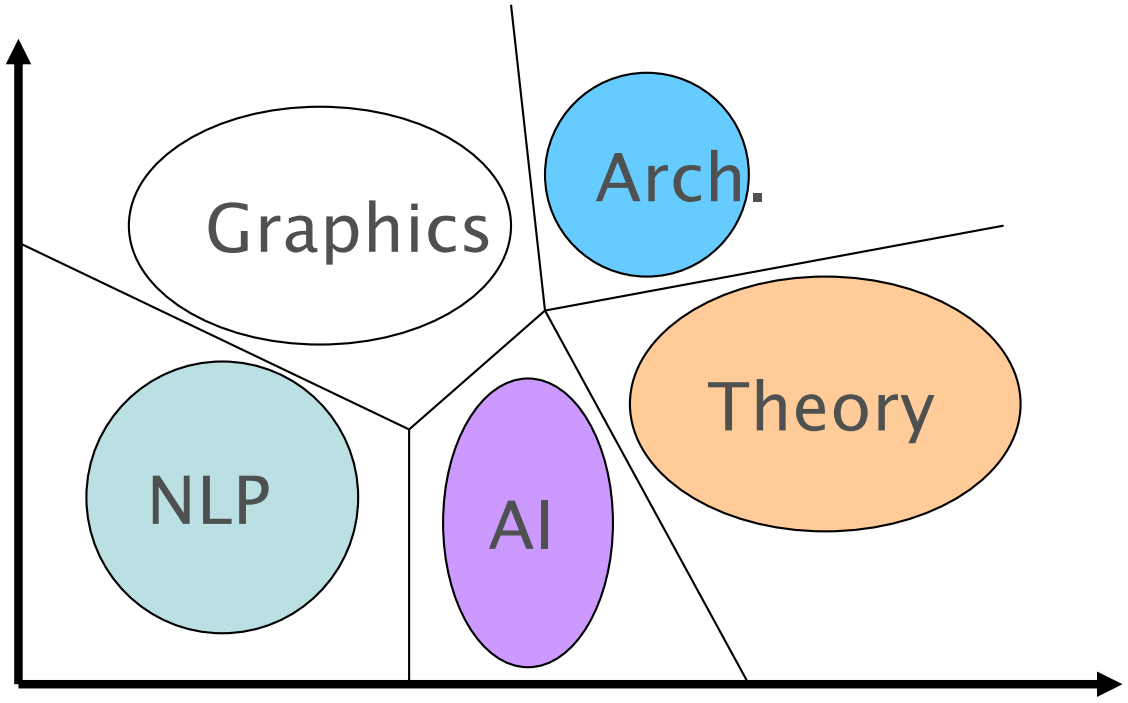
Determine

- The class of x: $c(x) \in C$, where $c(x)$ is a *classificaiton function* whose domain is X and whose range is C.
  - ***If x is spam then c(x)=1 otherwise c(x)=0***
  - ***We want to know how to build classification functions ("classifiers").***



A Graphical View of Text Classification

The feature space is partioned into regions of different classes, e.g. topics of research papers from computer science



Examples of Text Clasifications

- **CLASS LABELS=BINARY**
  - “spam” / “not spam”
- **CLASS LABELS=TOPICS**
  - “finance” / “sports” / “asia”
- **CLASS LABELS=OPINION**
  - “like” / “hate” / “neutral”
- **CLASS LABELS=AUTHOR**
  - “Shakespeare” / “Marlowe” / “Ben Jonson”

Indeed, Naïve Bayes is not the only approach one can images

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1. Manual classification

- Very accurate when job is done by experts
- Consistent when the problem size and team is small
- Difficult and expensive to scale

2. Automatic document classification

- Hand-coded rule-based systems
  - E.g., assign category if document contains a given Boolean combination of words
  - Accuracy can be high if a rule has been carefully refined over time by a subject expert also expert in the query language
  - Building and maintaining these rules is expensive



Classification Methods (2)

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3. Supervised learning

- All systems rely on machine learning
  - k-Nearest Neighbors (simple, powerful)
  - *Naive Bayes (simple method)*
  - Support-vector machines (more powerful)
  - Deep Learning (most powerful), using e.g. transformers for learning embedding (maybe using JEV)
  - … many other methods
- **No free lunch**: high quality performance may require hand-classified training data
- But data can be built up (and refined) by non-experts

How to get a classifier from the joint distribution / Bayesian network?

Reminder: Bayes' Rule

$$P(C,D) = P(C \mid D)P(D) = P(D \mid C)P(C)$$

C = Class, D = Document

$$P(C \mid D) = \frac{P(D \mid C)P(C)}{P(D)}$$

Maximum a posteriori Hypothesis

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$$c_{MAP} \equiv \underset{c \in C}{\operatorname{argmax}} P(c \mid D)$$

$$= \underset{c \in C}{\operatorname{argmax}} \frac{P(D \mid c) P(c)}{P(D)}$$

$$= \underset{c \in C}{\operatorname{argmax}} P(D \mid c) P(c)$$

Since P(D) is constant



Maximum Likelihood (ML) Hypothesis

If all hypotheses are a priori equally likely, we only need to consider the $P(D|c)$ term:

$$c_{ML} \equiv \underset{c \in C}{\operatorname{argmax}} P(D \mid c)$$

Naive Bayes Classifiers

Task: Classify a new instance D based on a tuple of attribute values $D = \langle x_1, x_2, \ldots, x_n \rangle$ into one of the classes $c_j \in C$

$$c_{MAP} = \operatorname*{argmax}_{c \in C} P(c \mid x_1, x_2, \ldots, x_n)$$

$$= \operatorname*{argmax}_{c \in C} \frac{P(x_1, x_2, \ldots, x_n \mid c) P(c)}{P(x_1, x_2, \ldots, x_n)}$$

$$= \operatorname*{argmax}_{c \in C} P(x_1, x_2, \ldots, x_n \mid c) P(c)$$

For instance, bag-of-word features when classifying text documents

Naïve Bayes (NB) Classifier: Assumption

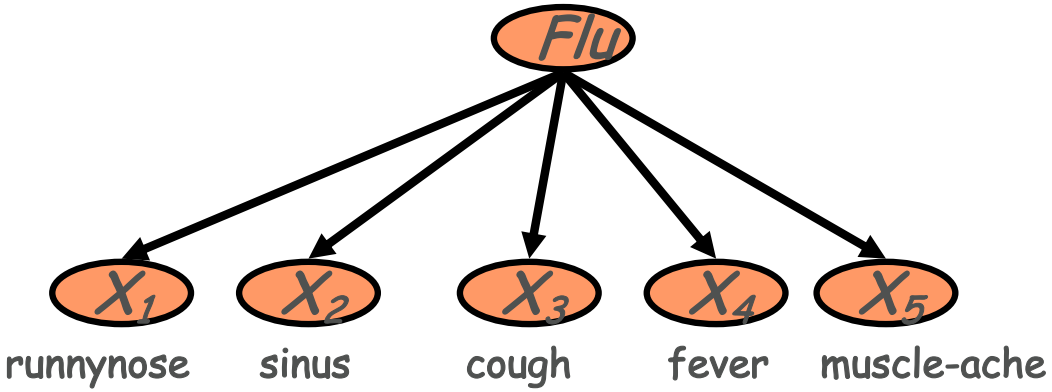
- $P(c_j)$
  - Can be estimated from the frequency of classes in the training examples.
- $P(x_1,x_2,\ldots,x_n|c_j)$
  - $O(|X|^n \bullet |C|)$ parameters
  - Could only be estimated if a very, very large number of training examples was available.

NB Conditional Independence Assumption to the rescue!

- ***Assume* that the probability of observing the attributes is equal to the product of the individual probabilities $P(x_i|c_j)$.**



The Naïve Bayes Classifier

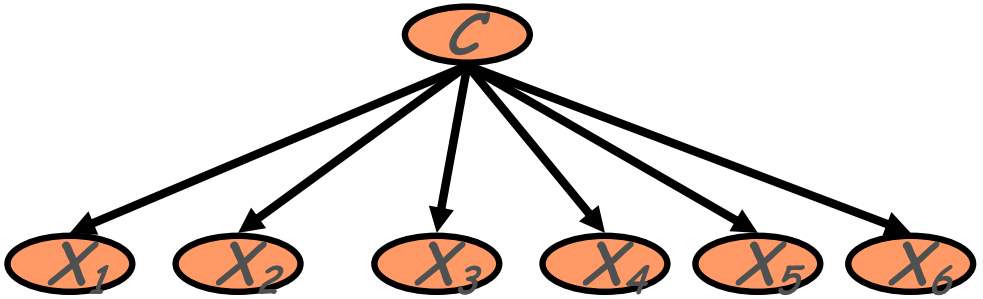


- Conditional Independence Assumption: features are independent of each other given the class:

$$P(X_1,\ldots,X_5 \mid C) = P(X_1 \mid C) \bullet P(X_2 \mid C) \bullet \cdots \bullet P(X_5 \mid C)$$

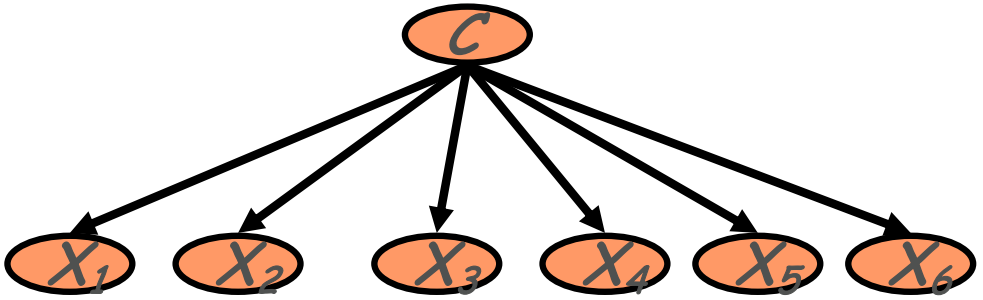
Learning the Model

Where do the parameters com from?



Learning the Model

Where do the parameters com from?



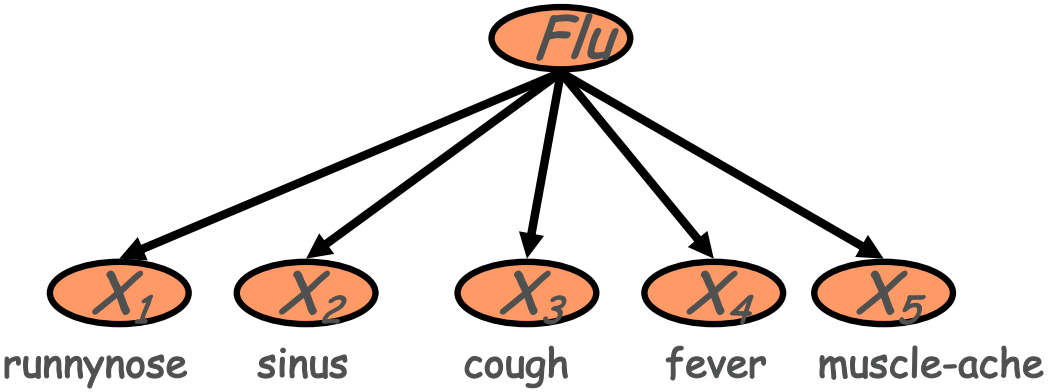
- First approach: maximum likelihood estimates
  - simply use the frequencies in the data

We will derive the equations later in class

$$\hat{P}(c_j) = \frac{N(C = c_j)}{N}$$

$$\hat{P}(x_i \mid c_j) = \frac{N(X_i = x_i, C = c_j)}{N(C = c_j)}$$

Problem with Max Likelihood



- What if we have seen no training cases where patient had flu and muscle aches?

$$P(X_1,\ldots,X_5 \mid C) = P(X_1 \mid C) \bullet P(X_2 \mid C) \bullet \cdots \bullet P(X_5 \mid C)$$

$$\hat{P}(X_5 = t \mid C = flu) = \frac{N(X_5 = t, C = flu)}{N(C = flu)} = 0$$

- Zero probabilities cannot be conditioned away, no matter the other evidence!

$$\ell = \arg\max_c \hat{P}(c) \prod_i \hat{P}(x_i \mid c) = 0$$

Smoothing to Avoid Overfitting

$$\hat{P}(x_i \mid c_j) = \frac{N(X_i = x_i, C = c_j) + 1}{N(C = c_j) + k}$$

We will derive the equations later in class

of values of X_i

- Somewhat more subtle version

overall fraction in data where $X_i = x_{i,k}$

$$\hat{P}(x_{i,k} \mid c_j) = \frac{N(X_i = x_{i,k}, C = c_j) + mp_{i,k}}{N(C = c_j) + m}$$

extent of "smoothing"

Wrap-up: (1) Naïve Bayes Learning

- From training corpus, extract *Vocabulary*
- Calculate required $P(c_j)$ and $P(x_k \mid c_j)$ terms
  - **For each c_j in C do**

$docs_j \leftarrow$ subset of documents for which the target class is c_j

$$P(c_j) \leftarrow \frac{|docs_j|}{|\text{total \# documents}|}$$

- **For each word x_k in *Vocabulary***

$n_k \leftarrow$ number of occurrences of x_k in all $docs_j$

$$P(x_k \mid c_j) \leftarrow \frac{n_k + 1}{|docs_j| + |Vocabulary|}$$

Wrap-up: (2) Naïve Bayes Classifying

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- Return c_{NB}, where

$$c_{NB} = \underset{c_j \in C}{\operatorname{argmax}} P(c_j) \prod_{i \in vocabulary} P(x_i \mid c_j)$$

- To avoid small numbers, use the sum of logs

Naïve Bayes works

- Uses a Naïve Bayes classifier on SPAMCOP
- EMail M is spam if P(Spam|M) > P(NonSpam|M)
- Method
  - Tokenize message using Porter Stemmer
  - Estimate P(W|C) using m-estimate (a form of smoothing)
  - Remove words that do not satisfy certain conditions
  - **Train: 160 spams, 466 non-spams**
  - **Test: 277 spams, 346 non-spams**
- Results: ERROR RATE of 4.33%
  - Worse results using trigrams

Take-Away Message:
Naive Bayes is Not So Naive

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- Naïve Bayes: First and Second place in KDD-CUP 1997 competition, among 16 (then) state of the art algorithms

  Goal: Financial services industry direct mail response prediction model: Predict if the recipient of mail will actually respond to the advertisement – 750,000 records.

- A good dependable baseline for text classification
  - But not the best!
- Optimal if the Independence Assumptions hold:
  - If assumed independence is correct, then it is the Bayes Optimal Classifier for problem
- Very Fast:
  - Learning with one pass over the data;
  - Testing linear in the number of attributes, and document collection size
- Low Storage requirements

Though deep networks are dominating today



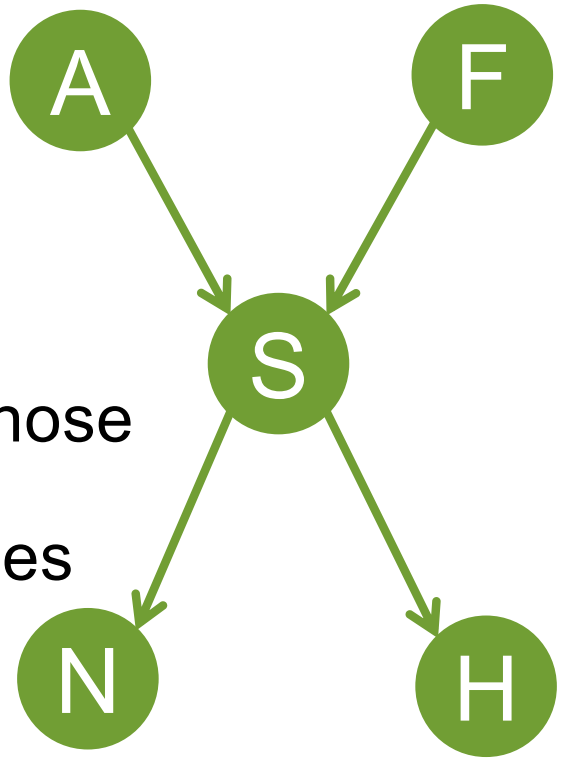
So far

- We've heard of Bayes nets
- We've played with Bayes nets
- We've even used them for classifying spam

- Now, we'll learn
  - the definition and semantics of BNs and
  - **relate them to independence assumptions encoded by the graph**

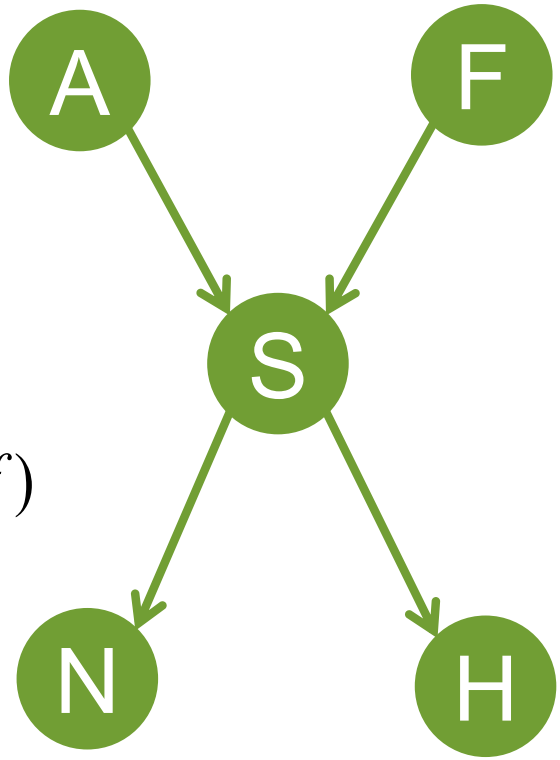
„Causal“ Structure

- Suppose we know the following:
  - The flu causes sinus inflammation
  - Allergies cause sinus inflammation
  - Sinus inflammation causes a runny nose
  - Sinus inflammation causes headaches
- How are these connected?



Possible Queries

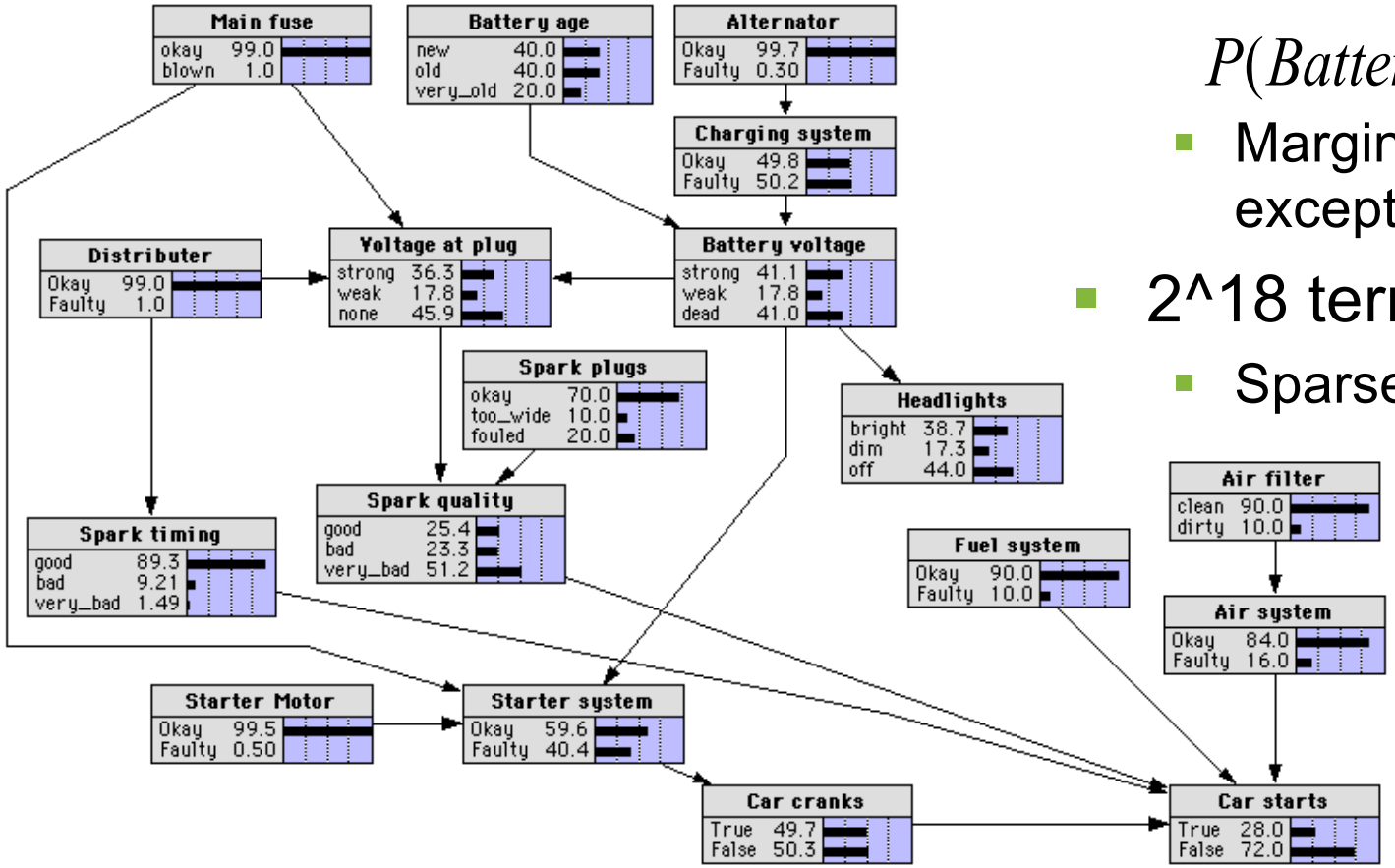
- **Inference**
  - Assume H=t, N=f. What is the probability P(A=t|H=t,N=f) of an allergic reaction?



- **Most probable explanation**
  - $\max_{f,a,s} P(F=f, A=a, S=s \mid H=t, N=f)$

- **Active data collection**
  - What is next best test, i.e., variable to observe

„Car Diagnosis“ Bayesian Network



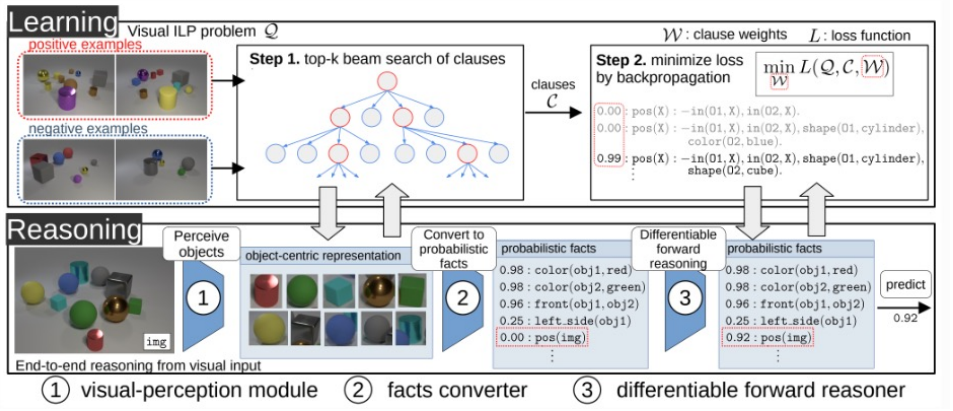
- 18 binary attributes
- Inference

$$P(BatteryAge \mid Starts = f)$$

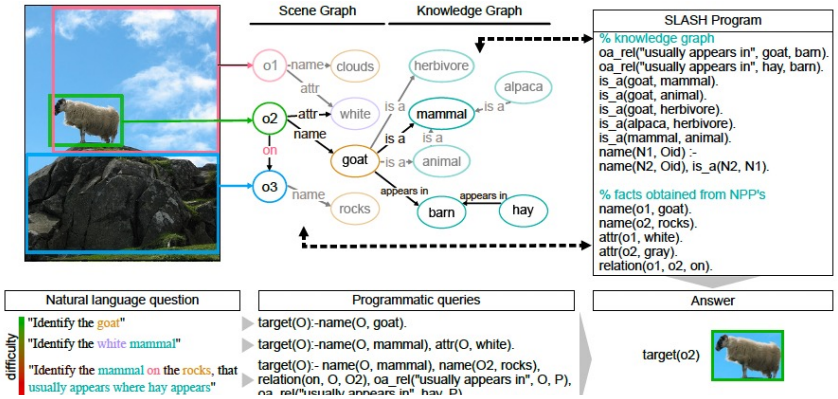
  - Marginalize all variables except BA and S
- 2^18 terms, why so fast?
  - Sparse structure !!!

Not impressed?

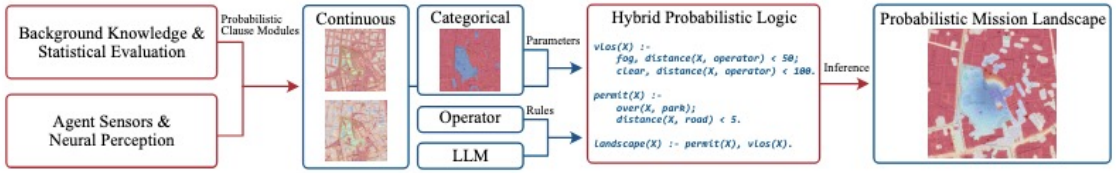
- We are currently dealing with (differentiable) models that have far more than 300 variables (nodes)
- > 2^300 ~ 10^{90} terms, btw, the age of the earth: ~ 4.54 × 10^{9} years ~ 10^{20} seconds



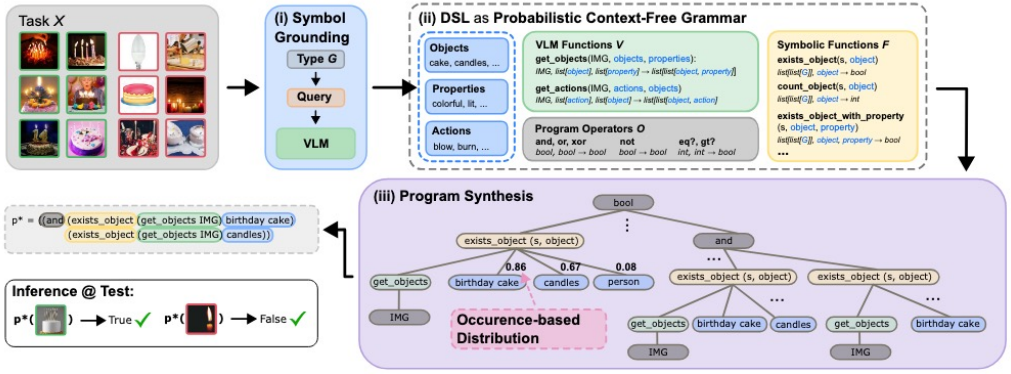
Shindo, Pfanschilling, Singh Dhami, Kersting (2023): **alphaILP: Thinking Visual Scenes as Differentiable Logic Programs.** *Machine Learning Journal (MLJ)*



Skryagin, Ochs, Singh Dhami, Kersting (2023): **Scalable Neural-Probabilistic Answer Set Programming.** *Journal of Artificial Intelligence Research (JAIR).*



Simon Kohaut, Benedict Flade, Daniel Ochs, Devendra Singh Dhami, Julian Eggert, Kristian Kersting (2025): **Probabilistic Mission Design for Neuro-Symbolic Unmanned Aircraft Systems.** *IEEE Transactions on Intelligent Transportation Systems:1-10.*

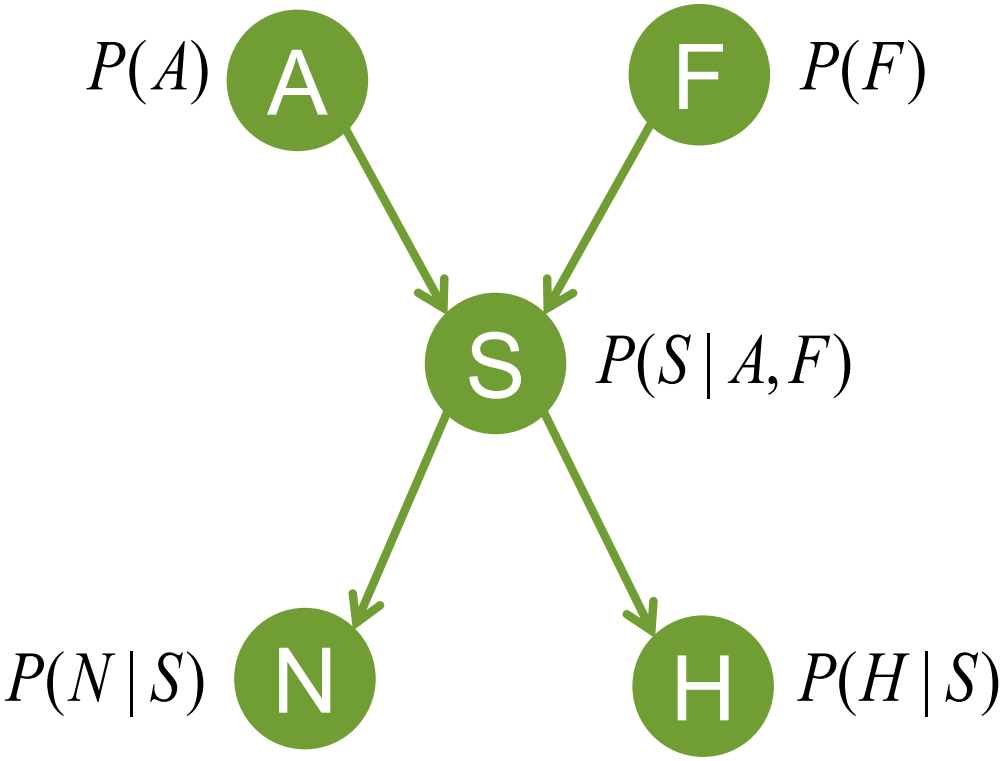


Wüst, Stammer, Shindo, Helff, Dhami, Kersting: **Synthesizing Visual Concepts as Vision-Language Programs.** CVPR 2026: 17346-17356

Factored Joint Distributions
Preview

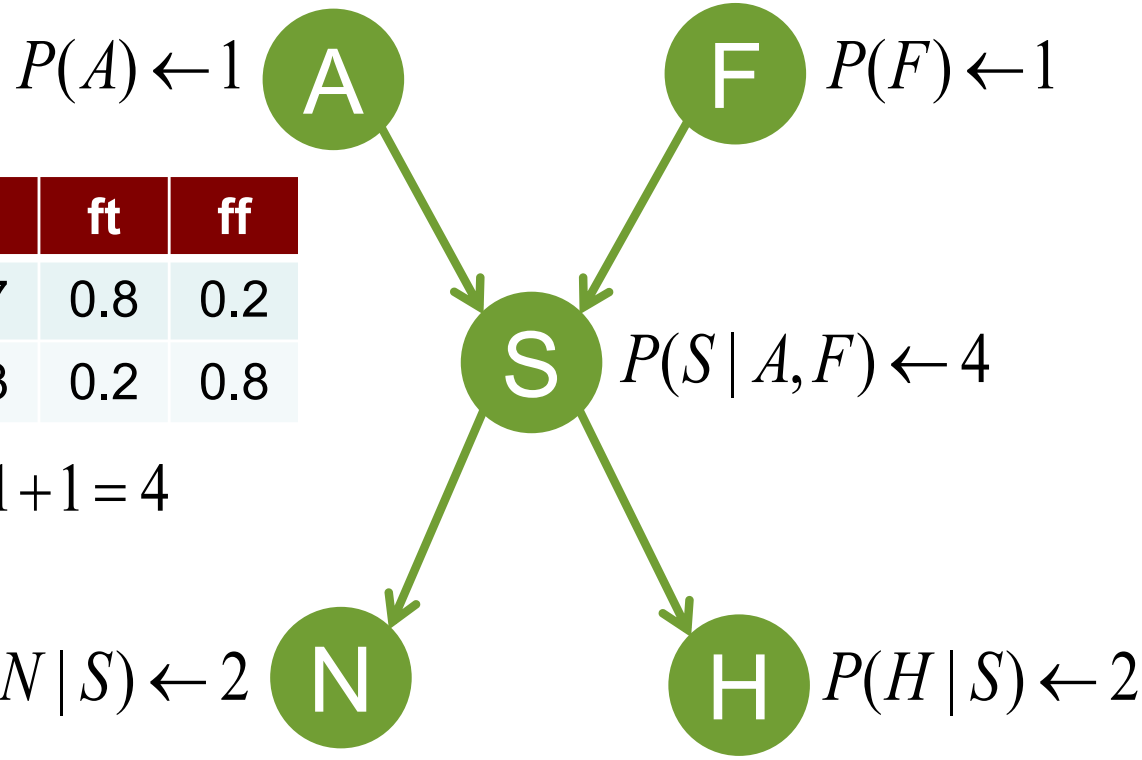
- Intuition: Nodes = Variables, Edges = Influences

$$
\begin{aligned}
P(A,F,S,H,N) &= P(A) \\
&\cdot P(F) \\
&\cdot P(S \mid A,F) \\
&\cdot P(N \mid S) \\
&\cdot P(H \mid S)
\end{aligned}
$$



Number of Parameters

- **Sparse structure**: compact representation for exponentially-large probability distributions
- Enumerative representation of binary variables: **2^5-1=31 parameters**



	A,F	tt	tf	ft	ff
S	t	0.9	0.7	0.8	0.2
	f	0.1	0.3	0.2	0.8

$$1+1+1+1=4$$

- **Using BNs**: **only 10 parameters**

Key: Independence Assumptions

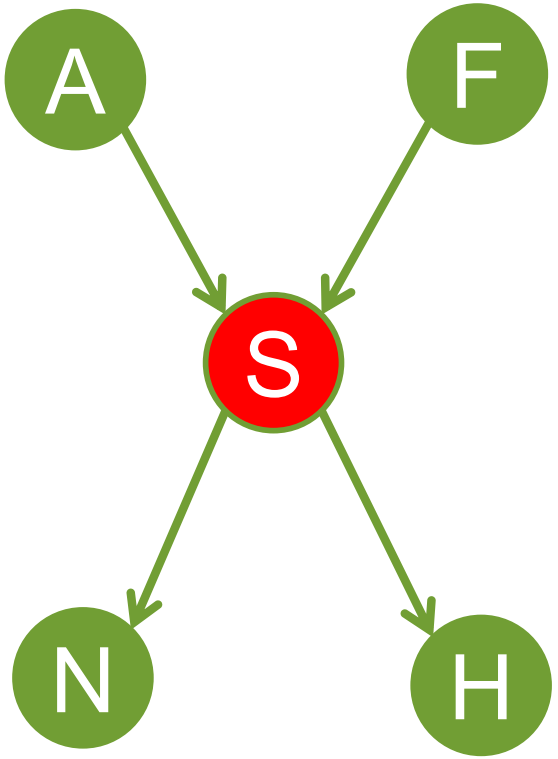
(intuitively clear)

$$\neg(F \perp H)$$

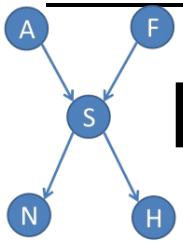
$$(F \perp H \mid S)$$

$$\neg(N \perp A)$$

$$(N \perp A \mid S)$$



Knowing sinus separates symptoms from causes



Recap: (Marginal) Independence

- Flu and Allergy are marginally independent

$$(A \perp F) \qquad P(A,F) = P(A) \cdot P(F)$$

A	t	f
t	0.3	0.7

F	t	f
t	0.1	0.9

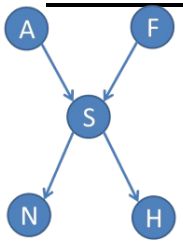
A,F	t	f
t	0.3*0.1=0.03	0.3*0.9=0.27
f	0.7*0.1=0.07	0.7*0.9=0.63

- General case:

$$\forall \text{ subsets } \mathbf{X}, \mathbf{Y} \subseteq \{X_1, \ldots, X_n\} : (\mathbf{X} \perp \mathbf{Y})$$

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} X_i$$





Recap: (Conditional) Independence

- Flu and Headache are (not) marginally independent
- Flu and headache are independent given Sinus infection

$$\neg(F \perp H) \qquad P(F,H) \neq P(F) \cdot P(H)$$

$$(F \perp H \mid S) \qquad P(F,H \mid S) = P(F \mid S) \cdot P(H \mid S)$$

$$P(F \mid H,S) = P(F \mid S)$$

- Generally: $(X_1 \perp X_2 \ldots X_n \mid C)$ iff

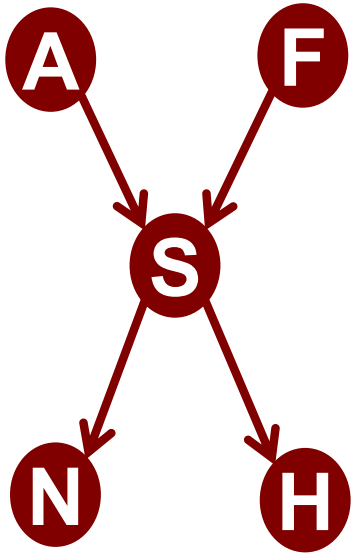
$$P(X_1, X_2, \ldots, X_n \mid C) = P(X_1 \mid C) \cdot P(X_2, \ldots, X_n \mid C)$$

$$P(X_1 \mid X_2, \ldots, X_n, C) = P(X_1 \mid C)$$



Local Markov Assumption

(Second most important slide!)



A variable X is independent of its non-descendants given its parents and only its parents **($X_i \perp \text{NonDescendants}_{X_i} \mid \text{Pa}_{X_i}$)**

$$\text{Pa}_F = \varnothing$$
$$\text{NonDescendants}_F = \{A\}$$
$$(F \perp A)$$

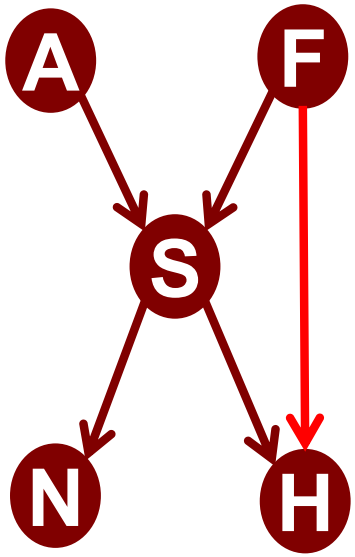
$$\text{Pa}_S = \{F, A\}$$
$$\text{NonDescendants}_S = \varnothing$$
$$(S \perp ?? \mid F, A)$$

NO ASSUMPTIONS

$$\text{Pa}_N = \{S\}$$
$$\text{NonDescendants}_N = \{F, A, H\}$$
$$(N \perp \{F, A, H\} \mid S)$$

How do we arrive at these independence statements?

Local Markov Assumption



A variable X is independent of its non-descendants given its parents and only its parents **($X_i \perp \text{NonDescendants}_{X_i} \mid \text{Pa}_{X_i}$)**

before edge included

$$\text{Pa}_H = \{S\}$$

$$\text{NonDescendants}_H = \{A, F, N\}$$

$$(H \perp \{A, F, N\} \mid S)$$

after edge included

$$\text{Pa}_H = \{F, S\}$$

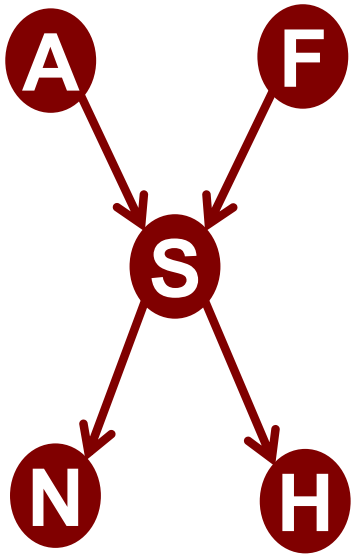
$$\text{NonDescendants}_S = \{A, N\}$$

$$(H \perp \{A, N\} \mid F, S)$$

How do we arrive at these independence statements?

Two independent events become conditionally dependent (negatively dependent) given that at least one of them occurs

Explaining Away / Berkson‘s Paradox



A variable X is independent of its non-descendants given its parents and only its parents **($X_i \perp NonDescendants_{X_i} \mid Pa_{X_i}$)**

$F \perp A$

S is not a parent but a descendant

$F \perp A \mid S$ is not implied by local Markov assumption

- Two causes „compete“ to „explain“ the observed data
- **Having a flu makes you less likely to have an allergy!!**
  P(A=t) <= P(A=t|S=t,F=T) <= P(A=t|S=t)

How do we arrive at these independence statements?



Explaining Away / Berkson‘s Paradox

Selection bias can induce a *false negative correlation* between independent traits.

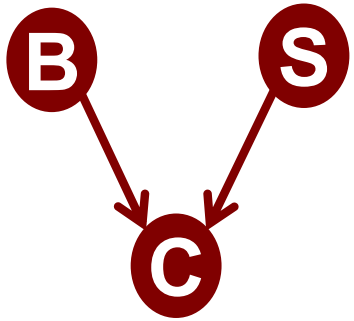


Example:
A school admits students who are *either* academically *or* athletically excellent.
In the overall population, the two abilities are independent.



But among admitted students (conditioned on admission), they appear **negatively correlated**, i.e., strong athletes tend to be weaker academically, and vice versa, purely due to the **selection rule**.

Explaining Away / Berkson‘s Paradox

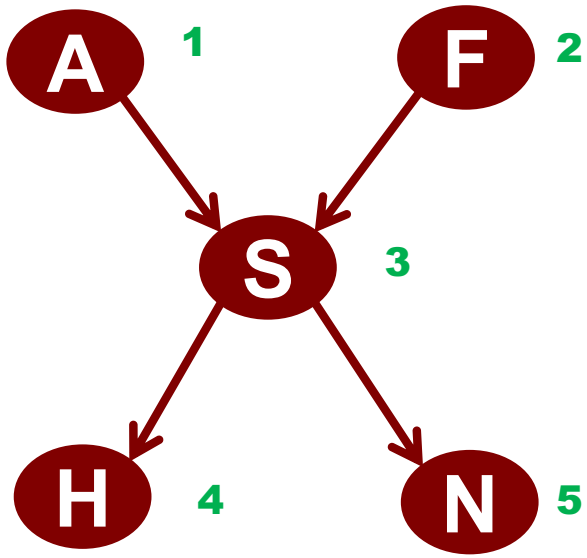


- A college which admits students who are either brainy or sporty (or both!)
- Two causes „compete“ to „explain“ the observed data

- Look at a population of college students
- **Being a brainy makes you less likely to be sportive and vice versa (if you are admitted)** because either property alone is sufficient to explain the evidence on C

$$P(S=t|C=t,B=T) <= P(S=t|C=t)$$

How to come up with the Joint Distribution



Consider **topological orders**

Now, to interpret a BN

1. Choose particular **chain rule order**
2. Apply independence assumption

$$P(A,F,S,H,N) = P(F)P(A)P(S \mid F,A)P(H \mid S)P(N \mid S)$$

$$P(A,F,S,H,N) = P(F)P(A \mid F)P(S \mid F,A)P(H \mid S,F,A)P(N \mid S,F,A,H)$$

$A \perp F$ $\quad P(A)$ $\quad P(S \mid F,A)$ $\quad H \perp \{F,A,N\} \mid S$ $\quad H \perp \{F,A\} \mid S$ $\quad P(H \mid S)$ $\quad N \perp \{F,A,H\} \mid S$ $\quad P(N \mid S)$

We can decompose due to the local Markov assumtion



Definition: Bayesian Network

(Most important slide!)

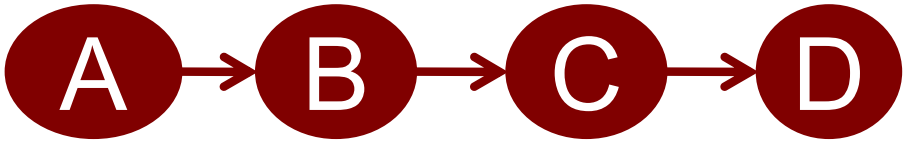


- Set of random variables $\{X_1,\ldots,X_n\}$
- Directed acyclic graph
  - loops are ok but **no directed cycles**
- CPT with each Xi: $P\left(X_i \mid \mathrm{Pa}\left(X_i\right)\right)$
- Joint distribution $P\left(X_1,\ldots,X_n\right)=\prod_{i=1}^{n} P\left(X_i \mid \mathrm{Pa}\left(X_i\right)\right)$
- **Local Makov assumption**

A variable X is independent of its non-descendants given its parents and only ist parents: **$(X_i \perp \mathrm{NonDescendants}_{X_i} \mid \mathrm{Pa}_{X_i})$**

Awesome, now we know exactly what Bayesian networks are! But there are still some representational questions left

- What distributions can be represented by a BN?
- What BNs can represent a distribution?
- What are the independence assumptions encoded, next to the ones due to the local Markov assumption



Local Markov assumption

$$\{A, B\} \perp D \mid C$$

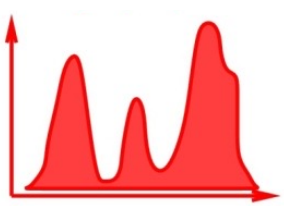
derived

$$A \perp D \mid C$$

Independencies in Real Problems

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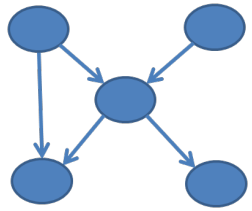
World, Data, Reality





Model, BN





True distribution P contains independence assertions

$$I(P)$$

Graph G encodes local independence assumptions

$$I_l(G)$$

Key representational assumption: $I_l(G) \subseteq I(P)$

The Representation Theorem

If conditional independencies in BN are a subset of conditional independencies in P, i.e., $I_l(G) \subseteq I(P)$

obtain

Then the joint probability distribution factorizes according to BN

$$P(X_1,\ldots,X_n) = \prod_{i=1}^{n} P(X_i \mid \mathrm{Pa}(X_i))$$

Important: Every P has at least on BN structure G

If joint probability distribution factorizes according to BN

$$P(X_1,\ldots,X_n) = \prod_{i=1}^{n} P(X_i \mid \mathrm{Pa}(X_i))$$

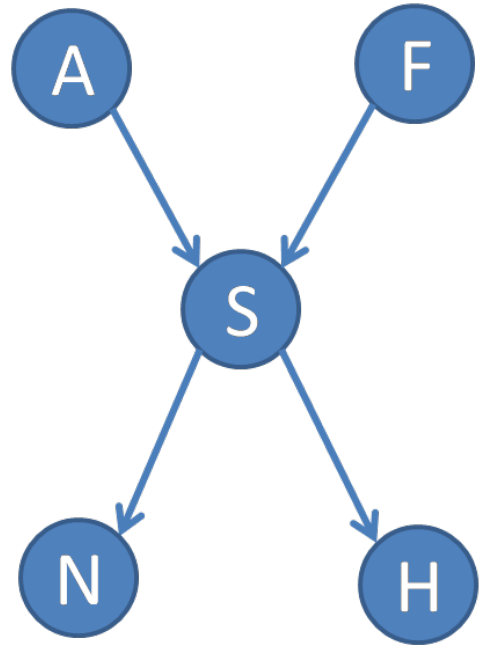
obtain

Then conditional independencies in BN are a subset of conditional independencies in P, i.e., $I_l(G) \subseteq I(P)$

Important: Read independencies of P from BN structure G

Local Markov Assumption & I-maps

A variable X is independent of its non-descendants given its parents and only its parents **(Xi ⊥ NonDescendants$_{Xi}$ | Pa$_{Xi}$)**

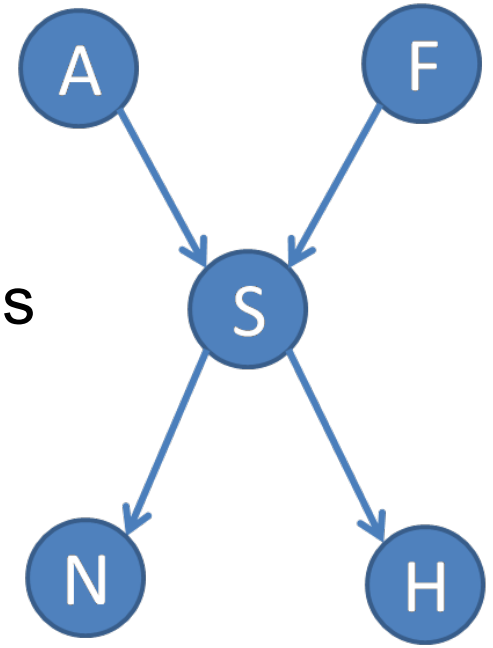


- Local independence assumption in BN structure G: $I_l(G)$
- Independence assertion of P: $I(P)$
- **BN structure G is an I-map (independence map) if:**

$$I_l(G) \subseteq I(P)$$

Factorized Distribution

- Given
  - Random variables $\{X_1,\ldots,X_n\}$
  - P distribution over the same variables
  - BN structure G over the same variables
- P factorizes according to G if



$$P(X_1,\ldots,X_n)=\prod_{i=1}^{n}P(X_i\mid \mathrm{Pa}(X_i))$$

The Representation Theorem

G is I-map of P

iff

P factorizes according to G

BN Rep. Theorem: I-Map to Factorization

G is I-map of P **obtain** P factorizes according to G

- Start with a topological ordering, wlog $X_1,\ldots,X_n$
- Apply chain rule

$$P\left(X_1,\ldots,X_n\right)=P\left(X_1\right)P\left(X_2\mid X_1\right)\ldots P\left(X_n\mid X_1,\ldots,X_{n-1}\right)$$

- Consider $P\left(X_i\mid X_1,\ldots,X_{i-1}\right)$
- We know that $\mathrm{Pa}\left(X_i\right)\subseteq\left\{X_1,\ldots,X_{i-1}\right\}$, i.e., there are no descendants of Xi in X1,…Xi-1
- Hence, due to local Markov assumption

$$P\left(X_i\mid X_1,\ldots,X_{i-1}\right)=P\left(X_i\mid\mathrm{Pa}\left(X_i\right)\right)$$



BN Rep. Theorem: I-Map to Factorization

P factorizes according to G **obtain** G is I-map of P

- Most likely done in the exercise session, ☺
- We have to show that the local Markov independence assumptions that hold in G also hold in P:

$$P\left(X_i|\mathrm{NonDes}\left(X_i\right),\mathrm{Pa}\left(X_i\right)\right)=P\left(X_i|\mathrm{Pa}\left(X_i\right)\right)$$

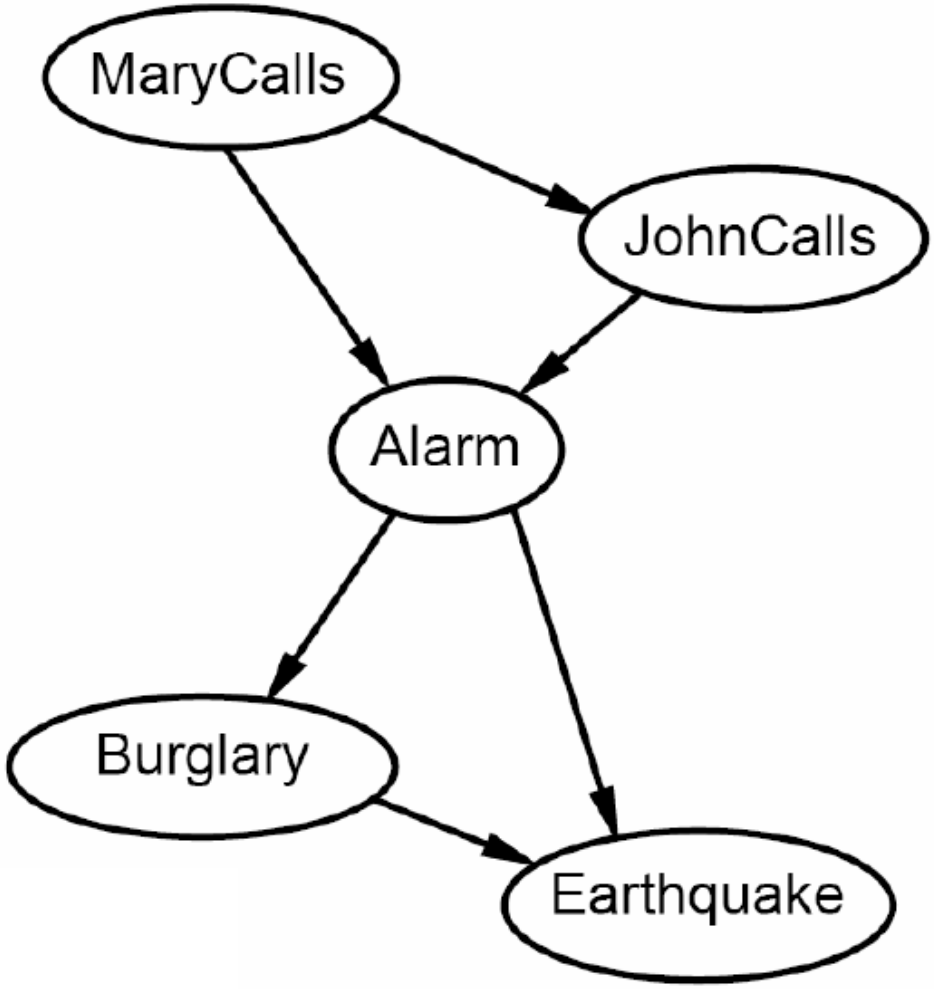
- Then apply definition of cond. prob. and simplify expressions

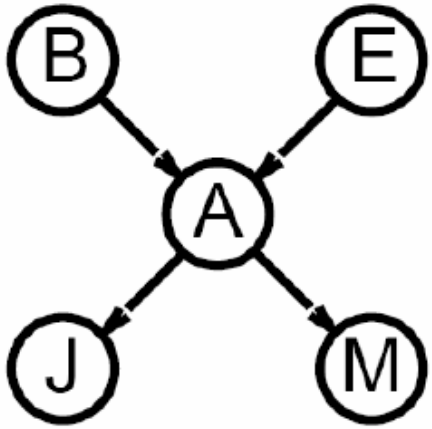


How do we come up with a Bayesian Network?

- **Given** a set of variables $\{X_1,\ldots,X_n\}$ and conditional indpendence assertions of P (estimated from data)
- **Choose** an ordering on variables, i.e., $X_1,\ldots,X_n$
- **For i = 1 to n**
  - Add Xi to the network
  - Define parents $\mathrm{Pa}(X_i)$ of Xi in graph as the minimal subset of $\{X_1,\ldots,X_{i-1}\}$ such that local Markov assumption holds
  - Define/learn CPT: $P(X_i \mid \mathrm{Pa}(X_i))$

Be careful! There are typically many Bayesian Networks for a joint distributions





This all depends on the ordering you are choising

Adding edges does not hurt

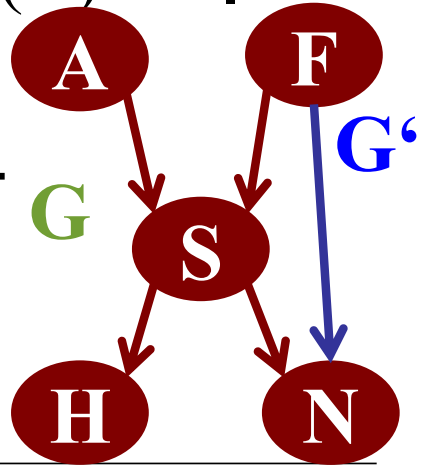
(Third most important slide!)

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- **Theorem:** Let G be an I-map for P, any DAG G' that includes the same directed edges as G is also an I-map for P.
  - **Corollary 1:** G' is strictly more expressive than G
  - **Corollary 2:** If G is an I-map for P, then adding edges still results in an I-map
- **Proof idea** for if $I_l(G) \subseteq I(P)$ then $I_l(G') \subseteq I(P)$.

  We show $I_l(G) \supset I_l(G')$ by induction

  Note that we are free to set the new CPT



$$P(X_i \mid Pa(X_i), X' = t) = P(X_i \mid Pa(X_i), X' = f)$$
$$= P(X_i \mid Pa(X_i))$$

What you need to know thus far

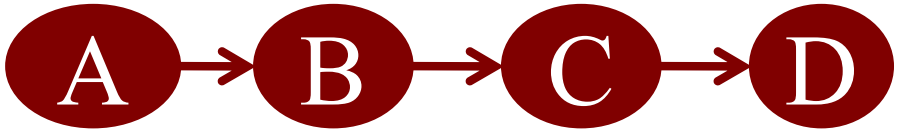
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- Independence and conditional independence
- Definition of a Bayesian network
- Local Markov assumption
- The representation theorem
  - G is I-map for P iff P factorizes according to G
  - Interpretation

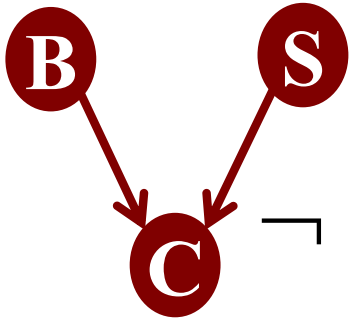


Independencies encoded in BN

- We said: all you need is the local Markov assumption **(Xi ⊥ NonDescendants$_{Xi}$ | Pa$_{Xi}$)**
- But then we talked about other (in)dependencies such as explaining away



$$A \perp D \mid C$$



$$B \perp S \qquad \neg\ B \perp S \mid C$$

- So, what are the independencies encoded by a BN?
  - Only assumption is local Markov but many other can be derived using the algebra of conditional independencies!